12-FA25-SP26-VI-WEB

Project Requirements and Specifications

Votegrity

12-FA25-SP26-VI-WEB
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I. Introduction

Votegrity is a growing company that provides voting services to HOA's, unions, and other groups. The more the company grows, the more daily elections can be helped. With this increase in elections, Votegrity has noticed increases in the difficulty of creating personalized election documents for each election. Their current method of creating election documents, using Microsoft Word [1], is slow and not intuitive for Votegrity staff.

Knowing this problem, we see opportunities to improve this election document workflow. Votegrity has given us the project of creating an intuitive drag-and-drop election document editor which will allow for a speed up in their election workflow. This will give them the ability to host an increased number of elections per day. By streamlining the placement of election document components (title, return address, voter address, etc.), assisted by hardcoded templates that match a certain document format (i.e., ballots, candidate statements, or notice documents), Votegrity will be able to expand their operations and increase profitability.

Additionally, this will be paired with more functionality such as mail-merge [1], which will be able to take a created election document template and a dataset of voter information and proceed to output hundreds or even thousands of personalized election documents. This will improve the election document workflow even more.

In the scenario where election document creation moves to an external Election Administrator in the future, having such a tool will only assist in this transition of responsibility, allowing whoever may create the election documents in the future to have a much easier time.

II. System Requirements Specification

II.1. Project Stakeholders

Our primary client is Votegrity. The major stakeholders in this project are internal Votegrity staff, contractors, and more generally, people that use internal tools. Accessibility and intuitive design will help ensure that these stakeholders have a good experience and can create election documents with ease for Votegrity elections. Other stakeholders include external customers that would purchase Votegrity voting services, and they need to be assured that election document quality is up to par. Another stakeholder is the voters themselves, as they will receive the documents our application is going to help create, so it is crucial this application is easy-to-use so internal Votegrity staff can ensure each election document has the required information for each voter.

Potential clients may include any company, organization, or person that needs a voting service. Votegrity is primarily used for HOA's, unions, or similar organizations.

II.2. Use Cases

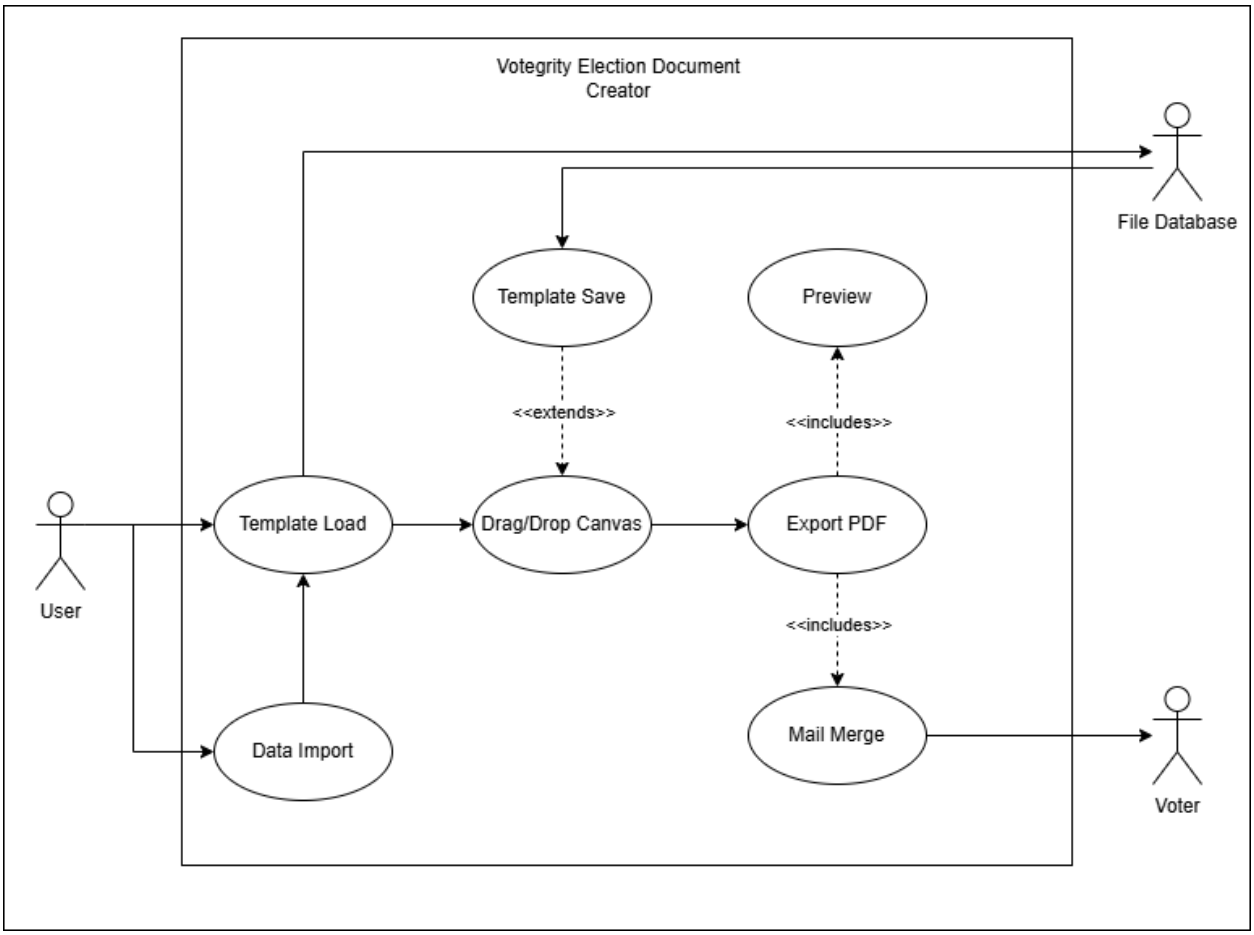

Fig. 1: Use case diagram for the Votegrity Election Document Creator showing the User, File Database, and Voter actors with core use cases (Template Load, Drag/Drop Canvas, Export PDF, Mail Merge, Data Import).

UC01: Template Load

Pre-Condition	-User is logged in
Post-Condition	-The template is loaded onto the canvas
Basic Path	1. The user initiates the Load Template function 2. The system retrieves and displays a list of templates from the database 3. The user selects a template 4. The system loads the selected template onto the canvas
Alternative Path	If the user chooses to not load a template, they can cancel the operation, and the canvas remains in its previous state.
Related Requirements	-[FR-3] Template Save/Load

UC02: Drag/Drop Canvas

Pre-Condition	-A template must be active on the canvas
Post-Condition	-The layout and content of the template are modified based on the user's actions.

Basic Path	1. The user selects a component from a tool menu 2. The user drags the component and drops it onto the canvas 3. The user can select, move and resize the component 4. The user can edit the properties and style of the selected component
Alternative Path	The user can undo/redo their actions
Related Requirements	- [FR-1] Drag-and-Drop Canvas , [FR-2] Layout Aids.

UC03: Data Import

Pre-Condition	- The user has a document template open on the canvas and has a dataset file containing voter information
Post-Condition	- The voter dataset is successfully parsed by the system, and its data fields are available for binding to template components.
Basic Path	1. The user selects the "Import Data" option. 2. The system prompts the user to upload a file. 3. The user selects the dataset file containing voter information. 4. The system reads the file and validates the data to prevent errors. 5. The system displays the field names from the dataset to the user.
Alternative Path	If the user selects a file with an unsupported format or corrupted data, the system displays an error message.
Related Requirements	- [FR-4] Data Import , [NFR-5] Data Integrity.

UC04: Template Save

Pre-Condition	- A template is currently being edited on the drag-and-drop canvas.
Post-Condition	- The current state of the template is saved as a file in the file database for future use.
Basic Path	1. The user initiates the "Save Template" function. 2. If the template is new, the system prompts the user to enter a file name. 3. The user provides a name and confirms the save action. 4. The system saves the template to the file database.
Alternative Path	The user cancels the save operation.
Related Requirements	- [FR-3] Template Save/Load , [NFR-3] Reliability.

UC05: Export PDF

Pre-Condition	- A template is finalized, and voter data has been imported (UC03) and mapped to the template (UC06).
Post-Condition	- A batch of personalized, print-ready PDF documents is generated and made available to the user.
Basic Path	1. The user initiates the "Export PDF" function. 2. The system first runs a pre-flight check to validate that all necessary information is present. 3. The system includes the Mail Merge process (UC06), using the imported data to populate the fields for each voter. 4. The system includes the Preview function (UC07), allowing the user to check a sample before final generation.

	5. The system generates all personalized documents as high-quality, printable PDFs.
Alternative Path	If the pre-flight check fails, the system notifies the user of the missing information so it can be corrected.
Related Requirements	- [FR-8] Batch Mail-Merge Generation, [FR-9] Export (Print-Ready PDF), [NFR-9] Export Quality.

UC06: Mail Merge

Pre-Condition	- A template is open on the canvas, and a voter dataset has been successfully imported via UC03.
Post-Condition	- The data fields from the imported dataset are mapped to their corresponding components on the template.
Basic Path	1. The user selects a component on the template meant to hold variable data (e.g., a text box for the voter's name). 2. The system presents the list of available field names from the imported data. 3. The user selects the appropriate data field to link to the component. 4. The system establishes the binding between the component and the data field. 5. The user repeats this process for all necessary components.
Alternative Path	The user can modify or remove an existing data binding.
Related Requirements	- [FR-5] Field Binding (Mail Merge).

UC07: Preview

Pre-Condition	- At least one data field has been bound to a template component via the Mail Merge process (UC06).
Post-Condition	- The user has visually confirmed that the data is appearing correctly in the template layout.
Basic Path	1. The user selects the "Preview" option. 2. The system generates a single sample document by populating the bound fields with data from one record. 3. The system displays this sample document to the user, ensuring that the preview is an exact representation of the final export. 4. The user reviews the document for correctness and formatting.
Alternative Path	The user can navigate through different records in the dataset to preview various resulting documents.
Related Requirements	- [FR-7] Preview of Merged Documents, [NFR-8] Accuracy.

II.3. User Stories

US-01:
As an election administrator, I want to drag and drop document components onto a canvas so that I can visually design the layout of our election mailers.

<u>Scenario 1: Positive Flow</u>

Given I have a new, blank document open

When I drag the “Voter Address Block” component from the toolbar onto the canvas
Then the address block component appears on the canvas where I dropped it.

Scenario 2: Negative/Edge Case
Given a component is already placed on the canvas
When I try to drag it outside the defined printable page boundaries
Then the component snaps back to the nearest edge within the boundary

US-02:
As a registered user, I want login so that I can access my previous templates and documents.

Scenario 1: Positive Flow
Given user has login credentials
When the user inputs valid credentials and clicks the login button
Then the user is logged into the system

Scenario 2: Negative
Given user has login credentials
When the user inputs the incorrect credentials and clicks the login button
Then the system displays an invalid credentials message

Scenario 3: Edge Case
Given user is a registered user
When user leaves either username, password, or both empty
And clicks the login button
Then the system will show invalid credentials.

US 03:
As a registered user I want to import pdf of a ballot, so that I can further add onto the document.

Scenario 1: Positive
Given user is registered
And Logins to the website
When the user clicks the import Icon
Then they will be prompted to select file to be dropped into editor
And can then add or remove items from the imported document.

Scenario 2: Invalid Document
Given user is register
And Logins to the website
When the user clicks the import Icon
Then they will be prompted to select file to be dropped into editor
When the user selects an unsupported file for our converter
Then the system will prompt user to provide a file type that the system can handle.

US-04:
As an election administrator, I want to link components on the canvas to specific fields from my imported data so that the correct voter information is on the correct component.

Scenario 1: Positive Flow
Given I have imported voter data and have a component on my canvas

When I select the component and bind it to the desired data field
Then the component on the canvas displays sample data from that field

Scenario 2: Negative Flow
Given I have a text component selected
When I try to bind it to a data field that does not exist in the imported source file
Then the system prevents the binding and displays a list of only the available data fields

US-05:
As a staff member, I want the system to run a validation check on my document so that I can identify and fix issues like overflowing text before generating the final files.

Scenario 1: Positive Flow
Given my document layout is complete and all data fields are bound correctly
When I click the “Validate Document” button
Then I receive a confirmation message

Scenario 2: Negative Flow
Given a voter record contains an exceptionally long name
When I run the validation check
Then the system flags the issues, stating where and what the problem is

US-06:
As an election administrator, I want to preview the document with actual voter data so that I can confirm the mail-merge is working correctly before starting the batch job.

Scenario 1: Positive Flow
Given a document template is linked to an imported voter list
When I enter preview mod and navigate to a specific record
Then the document on screen displays the voter data for the chosen record

Scenario 2: Negative Flow
Given the voter data for a specific record has a blank value for a field
When I preview the record
Then the document preview shows a blank space where the value should be

US-07:
As a staff member, I want to export the entire mail-merge batch as a single, print-ready PDF file so that I can easily send it to our printing service.

Scenario 1: Positive Flow
Given I have successfully generated a mail-merge for 5,000 voters
When I click the "Export as Collated PDF" button
Then the system generates and downloads a single PDF file containing 5,000 pages

Scenario 2: Negative Flow
Given the mail-merge generation process is running
When I close the browser tab accidentally
Then the export process is cancelled, and no incomplete file is created

US-08:

As a new staff member, I want the user interface to be intuitive and easy to navigate so that I can create an election document without needing extensive training

Scenario 1: Positive Flow

Given I am a new user who has just logged in for the first time
When I follow the on-screen prompts and icons
Then I can successfully create, populate, and preview a document in under 15 minutes

Scenario 2: Negative Flow

Given I am looking for the "Save Template" feature
When I cannot find it in the main "File" menu
Then I can use the search bar in the "Help" menu to locate the feature

US-09:
As an election official, I want to be confident that all uploaded voter data is encrypted and protected so that we comply with data privacy regulations and prevent unauthorized access

Scenario 1: Positive Flow

Given I am a logged-in, authorized user
When I access my project dashboard
Then I can view and manage the voter data I previously uploaded

Scenario 2: Negative Flow

Given an unauthorized person has the direct URL to my project's data file
When they try to access that URL without being logged into an authorized account
Then they are redirected to the login page and shown an "Access Denied" error

US-10:
As an election administrator, I want the system to automatically save my work and be consistently online so that I don't lose progress during a long design session or due to an unexpected outage

Scenario 1: Positive Flow

Given I am editing a document layout and have not manually saved in 5 minutes
When the system's auto-save feature is triggered
Then a small notification appears, and my changes persist

Scenario 2: Negative Flow

Given I was in the middle of a mail-merge generation when I lost my internet connection
When I log back into the system after my connection is restored
Then I see a notification asking if I want to resume the generation from where it left off

US-11:
As a staff member on a deadline, I want the system to process a large mail-merge of thousands of records efficiently so that I can generate and send the final files to the printer without a long delay

Scenario 1: Positive Flow

Given I have a validated template and a dataset of 10,000 voters
When I initiate the mail-merge generation process

Then the process completes in a reasonable time (e.g., under 10 minutes) and a progress bar accurately reflects the status

Scenario 2: Negative Flow

Given the system is processing a very large batch job (over 100,000 records)
When I navigate to another part of the application
Then the generation process continues to run in the background, and I can perform other tasks

US-12:
As an election administrator, I want to export a specific range of the generated documents so that I can easily handle a reprint for a specific precinct or correct a small batch of errors.

Scenario 1: Positive Flow

Given a mail-merge for 1,000 voters has been generated
When I choose the "Export to PDF" option, select "Export Range," and enter "100" to "200"
Then the system downloads a PDF file containing exactly 101 pages

Scenario 2: Negative Flow

Given a mail-merge for 1,000 voters has been generated
When I try to export a range from "900" to "1100"
Then the system displays a warning "The maximum record is 1000. Please enter a valid range." and does not start the export

US-13:
As a document designer, I want to be able to change the font, size, and style (bold, italics) of text components so that I can create a visually appealing and easy-to-read document that meets our branding guidelines.

Scenario 1: Positive Flow

Given I have a "Title" text component on the canvas
When I select the component, choose "Arial" font, set the size to "24pt," and click the "Bold" button
Then the text in the component updates immediately to reflect these changes

Scenario 2: Negative Flow

Given a text component is bound to a voter data field
When I apply bold styling to the entire component
Then the styling is applied correctly even when I preview different voter records with varying text lengths

US-14:
As a staff member, I want to be prompted to remap data fields when I load a template with a new data source so that I can use an old template even if my new CSV file has slightly different column names.

Scenario 1: Positive Flow

Given I load a template that expects a 'Voter_ID' field, but my new data source has 'VoterID' instead
When the system detects the mismatch

Then a dialog appears showing the unbound field 'Voter_ID' and suggests matching it to the 'VoterID' column from my new data source

Scenario 2: Negative Flow

Given I load a template and a new data source is missing a field entirely (e.g., no 'Polling_Location' column)
When I attempt to remap the fields
Then the system shows the 'Polling_Location' component as unbound, and I can either leave it blank or manually delete it from the canvas

II.4. Traceability Matrix

Functional Requirement	Use Cases	User Stories	Priority
FR-1: Drag-and-Drop Canvas	UC-02	US-01, US-08, US-10, US-13	0
FR-2: Layout Aids	UC-02	US-01, US-08, US-13	0
FR-3: Template Save/Load	UC-01, UC-04	US-02, US-03, US-14	0
FR-4: Data Import	UC-03	US-03, US-04, US-06, US-09, US-14	0
FR-5: Field Binding (Mail Merge)	UC-06	US-04, US-11	0
FR-6: Validation & Pre-flight Check	UC-05	US-05, US-07, US-09, US-10	1
FR-7: Preview of Merged Documents	UC-07	US-06, US-11, US-12	0
FR-8: Batch Mail-Merge Generation	UC-05	US-06, US-07, US-11	1
FR-9: Export (Print-Ready PDF)	UC-05	US-07, US-12	0

II.5. Functional Requirements

II.5.1. Template Builder

Functional Requirement	[FR-1] Drag-and-Drop Canvas
Description	The system will allow users to drag predefined components (Title, Address, Logo, Text) onto a canvas, move/resize them, and edit styles.
Source	Sponsor (Votegrity)
Priority	Level 0 (Essential and required functionality)
Functional Requirement	[FR-2] Layout Aids (Guides & Snapping)

Description	The system will provide alignment guides, snapping grids, and space-evenly different components to speed the election document process.
Source	Team discussion (key decision)
Priority	Level 1 (Desirable to aid election document creation process)
Functional Requirement	[FR-3] Template Save/Load
Description	The system will provide the option to save and load template to allow saving of documents and editing of previously saved documents.
Source	Sponsor (Votegrity)
Priority	Level 0 (Essential)

II.5.2. Data & Binding

Functional Requirement	[FR-4] Data Import
Description	The system will allow datasets of voter information to be read as field names, each entry corresponding to one voter.
Source	Sponsor (Votegrity)
Priority	Level 0 (Essential)
Functional Requirement	[FR-5] Field Binding (Mail Merge)
Description	The system will map fields to template components based on field names and the dataset imported.
Source	Sponsor (Votegrity)
Priority	Level 0 (Essential)
Functional Requirement	[FR-6] Validation & Pre-flight check
Description	The system will ensure the document information is filled correctly before final document generation.
Source	Team discussion (key decision)
Priority	Level 1 (Desirable)

II.5.3. Preview & Exporting

Functional Requirement	[FR-7] Preview of Merged Documents
Description	The system will allow the user to preview a test document with sample data to visually check that data is being placed in fields correctly.
Source	Team discussion (key decision)
Priority	Level 0 (Essential)
Functional Requirement	[FR-8] Batch Mail-Merge Generation

Description	The system will allow the generation of the personalized documents to be run in batches of N voters.
Source	Sponsor (Votegrity)
Priority	Level 1 (Desirable)
Functional Requirement	[FR-9] Export (Print-Ready PDF)
Description	The system will support the election documents created to be exported as directly printable PDF's.
Source	Sponsor (Votegrity)
Priority	Level 0 (Essential)

II.6. Non-Functional Requirements

II.6.1. Template Builder

Nonfunctional Requirement	Usability [NFR-1]
Description	The system should provide an intuitive and user-friendly interface that allows users with no specialized design skills to create and customize templates efficiently.
Nonfunctional Requirement	Performance [NFR-2]
Description	The system shall respond to user actions on the drag-and-drop canvas should be responsive and smooth
Nonfunctional Requirement	Reliability [NFR-3]
Description	The system shall autosave the current template state every set duration to prevent data loss in case of unexpected crashes or power failure.
Nonfunctional Requirement	Compatibility [NFR-4]
Description	The system should operate correctly on all major browsers (Chrome, Firefox, Edge, Safari) and adapt to various screen sizes.

II.6.2. Data & Binding

Nonfunctional Requirement	Data Integrity [NFR-5]
Description	The system shall validate imported datasets to prevent corrupted or malformed data from being used in document generation.
Nonfunctional Requirement	Security [NFR-6]
Description	The system shall securely handle user data, including encryption of sensitive voter information both in transit and at rest, to comply with privacy regulations.
Nonfunctional Requirement	Error Handling [NFR-7]
Description	The system shall provide clear error messages and guidance if data import or field binding fails or if required fields are missing.

II.6.3. Preview & Exporting

Nonfunctional Requirement	Accuracy [NFR-8]
Description	The system shall ensure that the previewed documents exactly represent how the final exported documents will appear, with no discrepancies in data placement or formatting.
Nonfunctional Requirement	Export Quality [NFR-9]
Descriptions	The system should generate print-ready PDF files that conform to standard PDF/A specifications to ensure compatibility with all commercial printing services.
Nonfunctional Requirement	Clarity [NFR-10]
Description	The system shall provide real-time feedback during export processes, including progress indicators and estimated completion times for batch jobs.

II.7. Standards

Standard / Code	Domain	Description	Application in Project
IEEE 830 – Software Requirements Specification	Software Documentation	Defines structure and clarity for software requirement documents.	Used to organize FR/NFR lists, user stories, and traceability to ensure consistency and testability.
W3C WCAG 2.2 – Web Content Accessibility Guidelines	Web / UI Accessibility	Ensures accessibility for users with visual or motor impairments.	Applied in interface design through text contrast, keyboard navigation, and assistive tool compatibility.
Washington Privacy Act	Data Protection / Legal Compliance	Establishes safeguards for collecting and handling personal data.	Ensures encryption, consent, and secure storage of voter and template data within the system.
ACM Code of Ethics	All	Promotes fairness, privacy, and responsible computing.	Guides ethical handling of election data and user information throughout system design.

III. System Evolution

Our current assumptions include, users are accessing the service via a website browser such as google, Firefox, or explore. From the web service they will have access to a file database of various election documents allowing them to upload and edit these templates for personal or election uses. A possible expansion of this user could be internal staff at Votegrity looking to

make changes to existing templates. We would then need to implement different privileges for separating users and roles in having us implement an extension of the user actor.

We plan on making use of several libraries to implement our features. Doing so creates reliance and dependency on these libraries that will either get updated or become outdated. We would then need to update our code base as libraries get updated, which may change how some functions may work, resulting in additional effort in updating our code. If libraries become outdated, we may need to replace or implement fixes to our code to accommodate the lack of updates from the library.

Currently we have planned for a few components such as textboxes, headers, or images that can be placed on the document. As we develop further, our scope for components may increase, resulting in more features to be added and thus, we need to be able to update all templates to accommodate these changes. As we continue to increase the components, we should consider the performance of the service so users will have seamless clean experience when using our product.

In addition, we plan on storing both templates from Votegrity and users so they may retrieve and save their work. Thus, we must implement proper safety and privacy for our user data. In doing so we will have to update and follow new guidelines as they are being implemented and published so our product follows proper procedures.

IV. Glossary

Defining technical terms used in the document.

Ballot: A document used to cast a vote in an election. In this context, a ballot may be a PDF or digital document customized for different elections.

Component: A draggable item used to build templates (e.g., text box, image/logo, address block, barcode, table).

Data Source: The uploaded voter dataset (CSV/XLSX) containing one row per voter.

Election Administrator: An individual (either internal or external to Votegrity) responsible for organizing and managing an election, including document creation.

Election Document: A formal document related to an election process, such as ballots, instructions, or registration forms.

Layout Aids: Editor features that improve placement (snap-to-grid, alignment guides, equal spacing).

Mail Merge: – The process that creates personalized documents by combining a template with each dataset record.

Preview: A rendered sample (or record-by-record viewer) showing exactly how the final PDF will look.

Record: A single voter’s row within the dataset.

Template: A reusable document layout (page size, margins, components) saved for future elections.

Autosave / Versioning: Periodic save with retrievable history to prevent work loss.

Votegrity: A secure online voting platform aiming to streamline the voting process through digital tools and services.

WYSIWYG: “What You See Is What You Get”; the editor aims to visually match the final exported output.

V. References

[1] Microsoft, “Use mail merge for bulk email, letters, labels, and envelopes,” Microsoft Support, https://support.microsoft.com/en-us/office/use-mail-merge-for-bulk-email-letters-labels-and-envelopes-f488ed5b-b849-4c11-9cff-932c49474705 (accessed Oct. 19, 2025).

[2] Microsoft, “Load or unload a template or add-in program,” Microsoft Support, https://support.microsoft.com/en-us/office/load-or-unload-a-template-or-add-in-program-2479fe53-f849-4394-88bb-2a6e2a39479d (accessed Oct. 19, 2025).

[3] R. Mambou, “How to generate a PDF file from a react component,” Medium, https://medium.com/@ryanmambou/how-to-generate-a-pdf-file-from-a-react-component-87707e869296 (accessed Oct. 19, 2025).

[4] Microsoft, “What is an online document editor?: Microsoft Word,” Microsoft, https://www.microsoft.com/en-us/microsoft-365/word/online-document-editing (accessed Oct. 19, 2025).

[5] “Documentation – Tailwind CSS,” Tailwind Labs, https://v2.tailwindcss.com/docs (accessed Oct. 19, 2025).

[6] “Washington Privacy Act (WPA),” Consumer Privacy Act, https://www.consumerprivacyact.com/washington/ (accessed Oct. 19, 2025).

[7] “1.3. Creating a Database,” PostgreSQL Documentation, https://www.postgresql.org/docs/current/tutorial-createdb.html (accessed Oct. 19, 2025).

[8] IEEE, IEEE Std 830-1998: IEEE Recommended Practice for Software Requirements Specifications, Institute of Electrical and Electronics Engineers, 1998.

[9] W3C, Web Content Accessibility Guidelines (WCAG) 2.2, World Wide Web Consortium, Oct. 2023. Available: https://www.w3.org/TR/WCAG22/

[10] ACM, ACM Code of Ethics and Professional Conduct, Association for Computing Machinery, 2020. Available: https://www.acm.org/code-of-ethics